A. Proposed Dataset

Research Project: Intelligent Planning and Geospatial Mapping of Document Acquisition Routes for Philippine Government IDs and Official Documents

Dataset working title: **PH-GovDocPath Dataset** | Generated from official .gov.ph sources

Proposed Dataset Name

Philippine Government Document Requirements and Acquisition Pathways Dataset (PH-GovDocPath Dataset)

Dataset Description

This dataset represents structured information about **Philippine government services, document requirements, office locations, and document dependencies** needed to obtain official government IDs and certificates (e.g., passport, National ID, NBI clearance, driver's license, professional license, SSS/PhilHealth/Pag-IBIG membership, voter registration, and business permits).

The dataset will contain:

- **Government services** — agency, service name, target document, transaction type, fees, processing time, and application method
- **Requirements** — required documents, acceptable IDs, supporting/conditional documents, and application procedures
- **Government offices** — office name, type, address, and contact details (latitude/longitude reserved for future geocoding)
- **Service locations** — which services can be applied for at which offices
- **Document dependencies** — prerequisite relationships between target documents and required documents (required, conditional, or alternative)

Relevance to the study: The proposed research focuses on intelligent planning and geospatial mapping of document acquisition routes. This dataset supports that goal by: (1) modeling what documents are needed for each government service; (2) modeling dependencies between documents; (3) linking services to application locations; (4) enabling association rule mining to discover common document combinations; and (5) supporting future route/path planning across agencies and offices.

All data is collected from **official Philippine government websites and Citizen's Charters**, not from third-party blogs or ready-made repositories.

Data Source

Data will be collected **only from official Philippine government websites** (.gov.ph domains) and official Citizen's Charters. No Kaggle, UCI, Wikipedia, or unofficial commercial sources will be used as primary data.

Agency	Website	Official URL(s)

PhilSys / PSA	philsys.gov.ph	https://philsys.gov.ph/registration-process/ ; https://philsys.gov.ph/supporting-documents/
PSA Civil Registration	psa.gov.ph	https://psa.gov.ph/civilregistration
DFA	consular.dfa.gov.ph	https://consular.dfa.gov.ph/passport/10-passport/61-passport-requirements-new-adult ; https://consular.dfa.gov.ph/valid-ids-for-passport-application/
NBI	nbi.gov.ph	https://nbi.gov.ph/citizens-charter/nbi-clearance-application/ ; https://nbi.gov.ph/how-to-apply-nbi-clearance/
LTO	lto.gov.ph	https://lto.gov.ph/ ; https://portal.lto.gov.ph/
PRC	prc.gov.ph	https://www.prc.gov.ph/list-of-requirements
COMELEC	comelec.gov.ph	https://www.comelec.gov.ph/?r=VoterRegistration/RegistrationProcedures
DTI	bnrs.dti.gov.ph	https://bnrs.dti.gov.ph/
SSS	sss.gov.ph	https://www.sss.gov.ph/become-an-sss-member/ ; https://www.sss.gov.ph/list-of-valid-ids/
PhilHealth	philhealth.gov.ph	https://www.philhealth.gov.ph/members/formal/registration.php
Pag-IBIG	pagibigfund.gov.ph	https://www.pagibigfund.gov.ph/membership/
GSIS	gsis.gov.ph	https://www.gsis.gov.ph/
Davao City LGU	davaocity.gov.ph	https://davaocity.gov.ph/services/processing-of-business-permits/

Every record preserves the **original source URL** and **date verified** (`last_verified`).

Data Collection Method

Primary method: Web scraping using a Python-based data collection pipeline.

1. **HTTP requests** (requests, cloudscraper) to fetch official HTML pages
2. **HTML parsing** (BeautifulSoup, lxml) to extract services, requirements, fees, procedures, and office information
3. **Playwright** (headless browser) only when necessary for Cloudflare-protected pages
4. **Post-processing** — deduplication, normalization, validation, export to five normalized CSV tables
5. **Manual verification / seeding** only when a site blocks access; labeled `scrape_status = seed` or `blocked`

Ethical constraints: Respect robots.txt and rate limits; no personal information; no fabricated fees or coordinates; empty values when unavailable.

Not used: surveys, Kaggle/UCI datasets, or unofficial blogs as primary sources.

Expected Number of Records

The final dataset contains **well over 500 records**, organized across five related tables:

Table	Records
Government services	36
Requirements	757
Government offices	13
Service locations	13
Document dependencies	679
Total data objects	**~1,498**

For data mining, the primary requirement-level dataset has **757 records**, with **727 live-scraped (success)** and **661 document-focused success records**. This exceeds the minimum of 500 records. Ready-made Kaggle/UCI datasets are **not** used.

> **Data quality note:** Approximately 96% of requirement records were live-scraped from official sources. Remaining rows are marked `seed` or `blocked` due to Cloudflare/WAF restrictions and are excluded from mining analysis unless manually verified.

Proposed Features / Attributes

1. government_services.csv

`service_id`, `agency`, `agency_acronym`, `service_name`, `target_document`, `transaction_type`, `applicant_type`, `application_method`, `appointment_required`, `processing_time`, `fee`, `source_url`, `source_title`, `last_verified`, `scrape_status`

2. requirements.csv

`requirement_id`, `service_id`, `requirement_name`, `requirement_name_normalized`, `requirement_type`, `requirement_role`, `conditional`, `condition`, `issuing_agency`, `document_format`, `quantity`, `source_url`, `source_section`, `requirement_description`, `last_verified`, `scrape_status`

3. government_offices.csv

`office_id`, `agency`, `agency_acronym`, `office_name`, `office_type`, `region`, `province`, `city`, `address`, `latitude`, `longitude`, `contact_number`, `source_url`, `last_verified`

4. service_locations.csv

`service_location_id`, `service_id`, `office_id`, `service_available`, `application_method`, `appointment_required`, `processing_time`, `fee`, `source_url`, `last_verified`

5. document_dependencies.csv

`dependency_id`, `service_id`, `target_document`, `required_document`, `required_document_normalized`, `relationship_type`, `condition`, `target_agency`, `required_document_agency`, `source_url`, `last_verified`

Key derived attributes: `relationship_type` (required, conditional, alternative); `requirement_role` (required_document, acceptable_id, supporting_document, procedure); `scrape_status` (success, partial, seed, blocked).

Proposed Data Mining Technique

Primary technique: Association Rule Mining (ARM)

Justification: Document acquisition involves discovering which documents frequently appear together across government services (e.g., PSA birth certificate + valid ID for passport). ARM treats each government service as a transaction and documents as items. Rules such as {PSA Birth Certificate, Valid ID} → {Passport Application} reveal common prerequisite patterns and support intelligent planning of document acquisition sequences.

Secondary technique (optional): Clustering or graph-based path planning for office locations once coordinates are geocoded.

Proposed Algorithms

For Association Rule Mining:

1. **Apriori Algorithm** — classic approach to find frequent itemsets and generate association rules
2. **FP-Growth (Frequent Pattern Growth)** — efficient FP-tree structure for larger requirement sets
3. **Eclat Algorithm** — vertical data format with depth-first search for sparse matrices

Proposed Performance Metrics

For Association Rule Mining:

1. **Support** — proportion of services containing a document itemset; measures how common a pattern is
2. **Confidence** — conditional probability that document B is required given document A; measures rule reliability
3. **Lift** — ratio of observed co-occurrence to expected; values > 1 indicate meaningful positive association

Repository: ph-gov-document-requirements-dataset | Output: data/final/ (5 CSV files)